QUANTITATIVE METHODS 1

Worksheet 4 – Bivariate distributions and regression

Detailed solutions

Exercise 1 – Detailed solution

$$\bar{x} = 3, \qquad \bar{y} = 6, \qquad V(x) = 2, \qquad V(y) = 4.8.$$

$$\mathrm{Cov}(x, y) = 3, \qquad r = \frac{3}{\sqrt{2 \times 4.8}} \approx 0.96825.$$

$$a = \frac{3}{2} = 1.5, \qquad b = 6 - 1.5(3) = 1.5.$$

$$\boxed{\hat{y} = 1.5x + 1.5}.$$

x_i	1	2	3	4	5
y_i	3	5	5	8	9
$\hat{y}_i$	3	4.5	6	7.5	9
e_i	0	0.5	-1	0.5	0

$$SSE = 1.5, \qquad SST = 24,$$

$$R^2 = 1 - \frac{1.5}{24} = 0.9375.$$

About 93.75% of the observed variation in Y around its mean is accounted for by the fitted linear model.

Exercise 2 – Detailed solution

$$\hat{y}(3.5) = 1.5(3.5) + 1.5 = 6.75.$$

Because $3.5 \in [1, 5]$, this is interpolation.

$$\hat{y}(10) = 16.5.$$

Since 10 lies far outside the observed range, this is risky extrapolation.

After adding $(10, 30)$, the new line is approximately

$$\hat{y} = 3.04918x - 2.70492.$$

The slope rises from 1.5 to about 3.05, showing that the new observation is highly influential. High correlation remains an association and cannot by itself establish causality.

Exercise 3 – Solution

$$A = \left(\frac{1+2+3}{3}, \frac{7+10+10}{3}\right) = (2, 9),$$

$$B = \left(\frac{4+5}{2}, \frac{14+14}{2}\right) = (4.5, 14).$$

$$a = \frac{14-9}{4.5-2} = 2, \qquad b = 5.$$

$$\boxed{y = 2x + 5}.$$

Exercise 4 – Solution

$$\widehat{X}_{10} = 2.155(10) + 9.429 \approx 30.98.$$

This exceeds the observed advertising maximum 27, so the first step extrapolates the time trend beyond month 8.

$$\widehat{Y}_{10} = 2.064(30.98) + 21.899 \approx 85.84.$$

The final sales forecast is about 85.84 thousand euros.